NANO EXPANSION BOARD
Documentation

Contact Us:
facebook.com/microbot.ph
microbot.ph@gmail.com

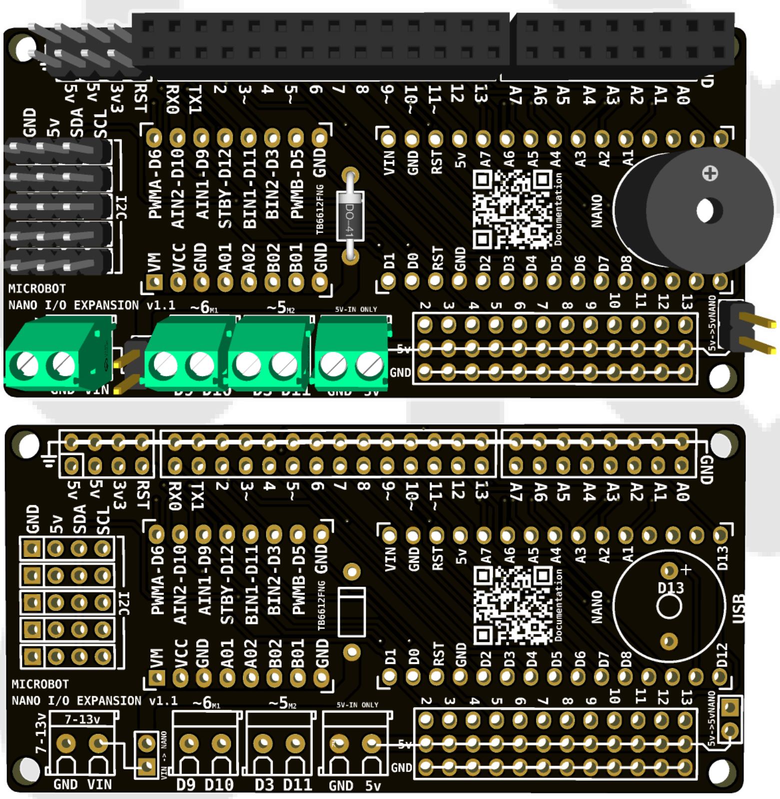
Overview:

The **Nano Expansion Board** is a versatile **Arduino Nano-compatible** development module designed for **robotics**, **automation**, and **embedded system** projects. It provides easy access to **GPIO** pins, **sensor** connections, and **motor/servo** control, making it ideal for rapid prototyping and educational applications.

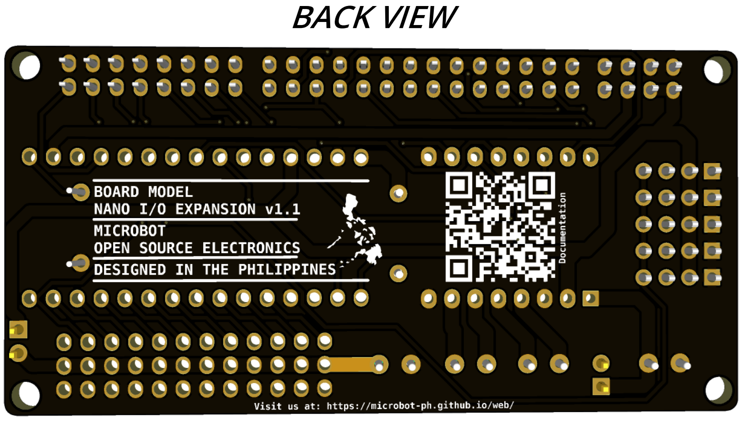
With dedicated mounts for the **Arduino Nano**, a **TB6612FNG** motor driver, and a **buzzer**, as well as multiple **I2C** and **servo** connections, this board centralizes control and power management while simplifying wiring and hardware integration.

Board Layout:

FRONT VIEW



BACK VIEW



Key Features:

- Integrated **TB6612FNG** Dual Motor Driver: Drive two DC motors with up to **1.2A** continuous current (**3.2A** peak) per channel, offering efficient and reliable performance for your robotics applications.
- Dedicated **Servo Motor Headers** (D2–D13): Easily connect up to **12 servo motors** directly to digital pins 2 through 13. Each header includes **ground** and **5V** rails for clean and organized power distribution.
- **Full Negative** (GND) Rails on All IO Pins: Simplify wiring and ensure stable grounding with dedicated negative rails across all input/output pins—ideal for sensor arrays and peripheral modules.
- Includes a dedicated pin for connecting a **Active buzzer**, enabling sound output for alerts or feedback.
- **Quint I2C Connectors**: Connect multiple I2C devices without the hassle of external splitters. With 5 dedicated I2C headers, you can easily chain displays, sensors, and other I2C-compatible modules.
- **Flexible Power Input Options**: Power the board through either the VIN (6–12V) or regulated 5V input. Note: Only one power source should be used at a time to prevent damage.

This expansion board is ideal for **beginners** and **experienced** makers alike, enabling quick prototyping and durable deployment in mobile robots, automation systems, and interactive projects.

Parts List:

QTY	Part Name
4	KF350 Terminal Block 2P
2	1x15 Female Header
2	1x8 Female Header
1	Jumper
1	1N4001 Diode
1	Buzzer
2	1x2 Male header
6	1x4 Male Header
2	1x14 Male Header
2	1x8 Male Header
3	1x12 Male header
1	PCB